

Learning Better Policies Through Safety Filter Distillation

Nathan Ludlow

Email: nludlow@andrew.cmu.edu

Abstract—In many robotics tasks, safety is not an auxiliary constraint but a prerequisite for task success: a robot that drives into a wall has not solved a navigation task. Hamilton–Jacobi (HJ) reachability shields and Control Barrier Functions (CBFs) encode precisely the structural information a learned policy requires to be both safe and high-performing: which states are unsafe, and what minimal deflection from a goal-directed action keeps the system clear of them. Yet they are conventionally attached to a policy as a runtime override that clips unsafe actions while leaving the underlying policy untouched. We argue this information is as useful for training a policy as for filtering one at runtime: distilling a safety filter into the policy lets it internalise “safe goal-directed action” rather than rediscovering it from collision penalties. During training, an analytic HJ filter labels each unsafe state with a convex combination of its evasive action and a goal-directed action, with the evasive weight increasing with depth into the unsafe set; a Soft Actor-Critic (SAC) student is then trained jointly against the task reward and a selective behavioural-cloning (BC) loss applied only at safety-critical states ($V \leq 0$). We evaluate this hypothesis on a Dubins-car navigation task. The distilled student attains 95.7% goal and 98.1% safe rates on hard-random scenes with limited sensor observations, significantly beating all baselines while matching or exceeding their safe rates.

I. INTRODUCTION & MOTIVATION

Safety is part of task success. A robot that drives into a wall has not solved a navigation task; a manipulator that crushes its workpiece has not assembled anything. Across surgical robotics, autonomous driving, and human–robot collaboration, the constraints (collision avoidance, force limits, dynamic feasibility) are not separable from the task: a controller that violates them has, by definition, failed at what it was deployed to do.

Safety filters carry structural, task-relevant information. HJ reachability shields [15] and CBFs [6] provide, given a state, both an indicator of whether that state is unsafe and a prescription for the minimal deflection from a goal-directed action that would keep the system clear of the unsafe set. This is precisely the structural knowledge that a learned policy struggles to acquire from a collision penalty alone: “do not crash” is a one-bit signal, whereas a filter encodes *where* to deflect and by *how much* at every state.

Idea: distill the filter into the policy. Filters are conventionally invoked at runtime as overrides that clip unsafe policy actions, so the policy never observes the filter’s reasoning during training and never internalises it. In this work we instead distill the filter during training: an analytic HJ filter generates a per-state action label that is goal-directed when the state is safely interior and that mixes in the filter’s evasive

action proportionally to how far below the BRT boundary the state lies. An SAC student is trained jointly against the task reward and a selective BC loss that is active only at safety-critical states ($V \leq 0$). The resulting policy can be deployed with or without the filter (the runtime overhead of the filter is negligible in either case), so the contribution we emphasise is the policy that respects safety as part of optimising the task, rather than the elimination of the runtime shield.

Key insight. A safety filter encodes the safe goal-directed action at every state, and distilling that information into a policy yields a policy that is better at the task than either the unshielded policy or the policy-with-runtime-filter combination achieves on its own.

Contributions. (1) An action-level filter-distillation method that augments standard SAC training with a behavioural-cloning loss gated on $V \leq 0$, so that the safety supervision is concentrated where it matters and is silent everywhere else. (2) An α -blended distillation label that interpolates between the filter’s evasive action and the goal-seeker proportionally to depth into the unsafe set, saturating at $V \leq -0.3$. (3) A multi-seed evaluation against seven baselines including SAFE-GIL [13], CBF-RL [25], curriculum filter distillation [1], and safety-augmented reward shaping: with privileged observations the distilled policy attains 100% goal and 100% safe rates, and with sensor-only observations it attains 95.7% goal and 98.1% safe rates, exceeding the next-best sensor-only baseline while matching or exceeding its safe rate.

II. RELATED WORK

We organise prior work according to *when* the safety filter participates in the system: at runtime, during training, or as a teacher distilled into the policy.

Runtime safety filtering. Shielded RL [4], HJ shields [15], CBFs combined with Gaussian-process dynamics [11], forward-reachable-set methods [20], and risk-budget CMDPs [14] all invoke the filter at runtime. Recovery RL [22] switches between distinct task and recovery policies. Such methods provide hard guarantees but leave the underlying policy unchanged: no information about the runtime correction propagates back into the policy parameters, so the policy never learns the filter’s notion of the safe goal-directed action.

Train-time filtering. Rollout filtering [18] maintains a collision-free replay buffer; CBF-RL [25] adds an actor regulariser that penalises actions the shield would correct; curriculum distillation [1] mixes in a state-uniform DAGger-style BC

loss with a time-decaying weight. These methods incorporate the filter into training but preserve only a coarse summary of its action-level information (a shielded transition or a scalar reward) or apply BC supervision uniformly across states.

Imitation, constrained RL, and reward shaping. SAFE-GIL [13] performs BC on clean expert labels collected under adversarial perturbations; we substitute the analytic HJ filter as a drop-in expert and report SAFE-GIL as a baseline. Constrained RL [5, 2, 21] and HJ-value-based shaping [16] encode the constraint as a reward term, but a scalar reward strips the action-level structure of the filter.

Learned safety representations. Conservative critics [10], Gaussian-process models [24, 23, 9], Lyapunov approaches [12, 7], and latent safety filters [17, 19, 3, 8] replace handcrafted filters with learned representations. Our use of a closed-form analytic teacher allows the experiments to isolate the contribution of distillation without confounding it with teacher approximation error.

In this work we distill the filter at the action level, apply BC supervision only at unsafe states, and soft-blend the filter’s evasive action with a goal-directed action. The student therefore receives a per-state label that is goal-aware wherever it can be and filter-aware wherever it has to be.

III. PROBLEM FORMULATION

We test two environment scenarios; in both we model the task as a CMDP [5] with state space $\mathcal{S}$, action space $\mathcal{A}$, dynamics P , task reward R , and discount γ . The privileged agent representation contains exact obstacle geometry $\{(o_x^i, o_y^i, r^i)\}_{i=1}^N$; the sensor-based agent observation comprises only an ego-frame goal vector and body-frame raycast distances.

Privileged HJ filter. For a Dubins car with speed v , maximum turn rate $u_{\max}$, and turning radius $\rho = v/u_{\max}$,

$$V_i(s) = \max(\|C_L - o_i\|, \|C_R - o_i\|) - (r^i + \rho),$$

where $C_L = (x - \rho \sin \theta, y + \rho \cos \theta)$ and $C_R = (x + \rho \sin \theta, y - \rho \cos \theta)$ are the two limit-cycle centres. The multi-obstacle safety value is $V(s) = \min_i V_i(s)$, and $V \leq 0$ delineates the BRT, the set from which no constant-input evasive trajectory remains collision-free. The closed-form expression requires no optimisation, gradients, or learned approximation, which motivates the Dubins setting as a controlled testbed for studying distillation.

Filter interfaces. We use two interfaces of the filter. The *evasive action* $u_{\text{filter}}(s) = \arg \max_u V(s')$ maximises the next-state safety value over a dense action grid and is goal-agnostic. The *action projection* $u_{\text{proj}}(s) = \arg \min_u |u - u_{\text{goal}}|$ subject to $V(s') > \epsilon$ returns the closest action to a privileged goal-seeker u_{goal} that keeps the next state safely inside the safe set, falling back to $\arg \max_u V$ if no candidate satisfies the constraint.

Two label families. The student is trained against one of two filter-derived labels:

- **PROJ (hard projection):** $u_{\text{teacher}} = u_{\text{proj}}$.
- **BLEND (α -blend):** $u_{\text{teacher}} = \alpha u_{\text{filter}} + (1 - \alpha) u_{\text{goal}}$ with $\alpha = \text{clip}(-V/c, 0, 1)$ and $c = 0.3$.

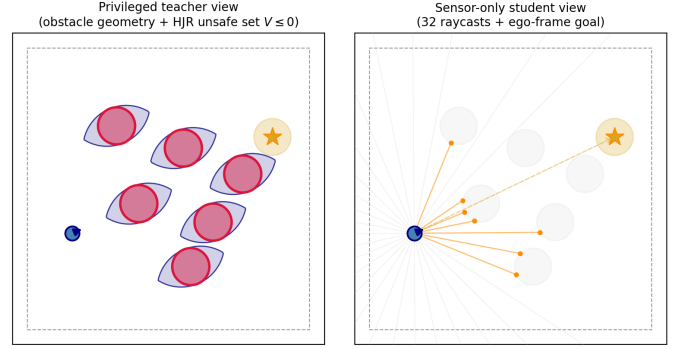


Fig. 1. Privileged-information distillation framing. **Left:** the teacher observes the agent, the goal, and exact obstacle geometry; the navy lens-shaped regions are the HJR unsafe set $\{V \leq 0\}$ at the agent’s current heading. **Right:** the student observes only ego-frame goal information and 32 body-frame raycast distances; obstacles are faded to indicate that they are present in the world but not in the student’s input.

Justification of the weighting. The blend weight α controls how much the teacher trades goal direction for safety, and we tie it directly to how unsafe the state is. When the state is well inside the safe set ($V \gg 0$), $\alpha = 0$ and the teacher returns the goal-directed action. As V drops below zero, α rises linearly with $-V$, mixing more of the filter’s evasive action into the label. Once $V \leq -c$, α saturates at 1 and the teacher returns the pure evasive action. How deep the state is in the unsafe set therefore decides how much the policy must give up goal direction for safety. The projection label, by contrast, is bang-bang in V : goal-directed up to the boundary, then $\arg \max_u V$ as soon as the state crosses it. The blended label gives a smooth signal across V , which a stochastic SAC actor can fit more easily.

IV. PROPOSED APPROACH

The pipeline reduces to a single per-state decision during training: at safety-uncritical states the policy is supervised only by the SAC objective and learns from reward, whereas at safety-critical states ($V \leq 0$, or where the projection differs from the goal-seeker by more than a small threshold) the policy additionally receives a BC loss against the α -blended label. This selective gating keeps the BC anchor informative where it matters and silent where it does not, and is the key distinction from curriculum-style schemes that apply BC supervision uniformly across states.

Environment. We implement a Dubins-car goal-navigation environment that supports 3–10 obstacles and configurable termination semantics. Scenes are generated randomly and then verified for navigability with a BFS check on a 40×40 grid. At training time, both collision termination and out-of-bounds termination are disabled, so that the agent receives full-episode rollouts rather than resetting after a single violation.

Sensor-only observation (35-dimensional). $[d_{\text{goal}}, \sin \alpha_{\text{body}}, \cos \alpha_{\text{body}}, r_0, \dots, r_{31}]$, where α_{body} is the goal direction in the body frame and $\{r_k\}$ are raycast distances at uniformly spaced bearings.

Reward. $r_t = -d_{\text{goal}} + 500 \cdot \mathbf{1}[\text{goal}] - 300 \cdot \mathbf{1}[\text{enter obstacle}]$. The collision penalty is one-shot on the outside-to-inside transition rather than per-step inside the unsafe set, capping the worst-case episode penalty so that it does not dominate the value function and induce a “stay still” policy.

Training objective. Training is standard SAC augmented with the selective BC loss. At every gradient step the actor minimises

$$\mathcal{L}_{\text{actor}} = E[\alpha_{\text{SAC}} \log \pi_{\theta}(a | o) - Q(s, a)] + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}(\theta),$$

where $\mathcal{L}_{\text{BC}}(\theta) = E_{s \in \mathcal{S}_{\text{crit}}}[\|\pi_{\theta}(o(s)) - u_{\text{teacher}}(s)\|^2]$ is the per-state mean-squared error between the actor output and the teacher label, restricted to the safety-critical set

$$\mathcal{S}_{\text{crit}} = \{s : V(s) \leq 0 \text{ or } |u_{\text{proj}}(s) - u_{\text{goal}}(s)| > 0.02\}.$$

Approximately 1–5 % of states are in $\mathcal{S}_{\text{crit}}$ in steady state; we use $\lambda_{\text{BC}} = 50$, which makes the BC term comparable in scale to the SAC actor objective at active states. The BC term is the only component that injects filter information into the policy, and it is active only at safety-critical states.

Baselines. We compare against pure SAC (no BC term), SAC with a runtime filter that overrides unsafe actions, and four learning baselines. *Curriculum Filter Distillation* [1] trains in two phases: a pure BC phase on filter labels followed by a pure RL phase, with no joint optimisation. *Safety-Augmented Reward Shaping* keeps the SAC objective unchanged but adds a penalty term that fires whenever $V(s) \leq 0$, encoding the filter as a scalar reward signal rather than an action-level label. *SAFE-GIL* [13] substitutes the analytic filter for SAFE-GIL’s human expert, executes the adversarial mixture $u_{\text{exec}} = (1 - \eta) u_{\text{base}} + \eta u_{\text{adv}}$ with $u_{\text{adv}} = \arg \min_u V(s')$ and $\eta = 0.3$, and trains via pure BC on clean expert labels. *CBF-RL* [25] projects each policy proposal onto the safe set, executes the projected action, and adds a CBF regulariser $\lambda_{\text{CBF}} E[\|\pi_{\theta}(o) - u_{\text{safe}}\|^2]$ to the actor loss; at evaluation it is deployed without a runtime shield. Finally, *SAC + BC (Evasive Filter Distillation)* is an ablation of our method that uses the filter’s evasive action as the teacher label at all states (rather than the α -blend or projection), and provides a direct test of whether the action-level structure in our two label families is what drives the gains.

V. RESULTS

All experiments use a random number of obstacles $n_{\text{obs}} \in [3, 10]$. Training and evaluation scenarios are sampled disjointly, so evaluation environments are unseen during training. Each method is trained with three random seeds and evaluated on the same 1,000 random scenes. We report results in two settings: a *privileged* variant in which the agent observes exact obstacle geometry, and a *sensor-only* variant in which the agent observes only an ego-frame goal vector and 32 raycast distances.

A. Privileged-View Results

Table I reports the privileged-view results.

TABLE I
PRIVILEGED-VIEW RESULTS ON HARD RANDOM ($n = 1000$, MEAN $\pm$ STD OVER THREE SEEDS).

Method	Goal	Safe
SAC	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
SAC w/ Runtime filtering	0.0 $\pm$ 0.0	100.0 $\pm$ 0.0
Curriculum Filter Distillation (CFD)	13.0 $\pm$ 5.6	42.0 $\pm$ 2.1
Safety Reward Shaping (SARS)	43.4 $\pm$ 4.3	91.0 $\pm$ 2.6
SAFE-GIL	47.3 $\pm$ 3.5	100.0 $\pm$ 0.0
CBF-RL	36.0 $\pm$ 3.0	100.0 $\pm$ 0.0
SAC + BC EVASIVE	23.0 $\pm$ 4.7	100.0 $\pm$ 0.0
SAC + BC PROJ (Ours)	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
SAC + BC BLEND (Ours)	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0

We draw four observations from Table I. *First, pure SAC fails completely.* Without any safety supervision the policy never escapes the early collision regime: at 0% goal and 0% safe it crashes on every evaluation scene. Adding a runtime filter recovers safety (100%) but does not recover the task: the shielded SAC reaches the goal in 0% of episodes, because the runtime override prevents collisions but propagates no information about the safe goal-directed action back into the policy. This pair of rows establishes that safety and task return are coupled: the filter alone is not sufficient to make a learned policy task-competent. *Second, the existing learning baselines occupy an intermediate regime.* Curriculum Filter Distillation, Safety-Augmented Reward Shaping, SAFE-GIL, and CBF-RL all reach goal rates between 13% and 47%, with safe rates ranging from 42% (Curriculum) to 100% (SAFE-GIL, CBF-RL). These methods incorporate the filter at training time but only at the level of a scalar reward, a state-uniform BC anchor, or a regulariser on shielded transitions; the action-level structure of the filter is not preserved in the supervision signal. *Third, the SAC + BC (Evasive Filter Distillation) ablation, which uses the filter’s evasive action as a uniform teacher, attains 23% goal at 100% safe.* This is the closest sibling of our method, identical except that it omits the goal-aware α -blend or projection structure. Its low goal rate isolates the contribution of the goal-aware label: the evasive action alone teaches safety but not goal-reaching. *Fourth, both proposed variants saturate at 100% goal and 100% safe.* The action-level supervision combined with a goal-aware label is what closes the gap; the next-best learning baseline (SAFE-GIL) trails by 53 percentage points on goal rate.

B. Sensor-Only Results

A privileged shield is not a deployable assumption: a real robot has access only to its onboard sensors. Table II repeats the comparison with the runtime filter disabled; the agent’s observation now consists of its state, 32 raycasts, and an ego-frame goal vector.

Figure 2 visualises the gap. Our two proposed variants are the only methods that retain near-privileged performance under the realistic sensor model: BLEND attains 95.7% goal at 98.1% safe, and PROJ attains 90.3% goal at 98.2% safe. Every other method either loses substantial goal rate when geometry

TABLE II

SENSOR-ONLY RESULTS ON HARD RANDOM ($n = 1000$, MEAN $\pm$ STD OVER THREE SEEDS). NO RUNTIME SHIELD IS ACTIVE.

Method	Goal	Safe
SAC	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Curriculum Filter Distillation (CFD)	5.0 $\pm$ 5.6	21.0 $\pm$ 5.7
Safety Reward Shaping (SARS)	13.5 $\pm$ 7.3	42.7 $\pm$ 7.1
SAFE-GIL	36.6 $\pm$ 5.5	82.3 $\pm$ 4.3
CBF-RL	32.0 $\pm$ 5.2	92.2 $\pm$ 2.0
SAC + BC EVASIVE	12.0 $\pm$ 3.6	97.4 $\pm$ 0.9
SAC + BC PROJ	90.3 $\pm$ 2.1	98.2 $\pm$ 1.0
SAC + BC BLEND	95.7 $\pm$ 0.8	98.1 $\pm$ 1.1

TABLE III

REWARD-SHAPING ABLATION ON HARD RANDOM ($n = 50$, OVER THREE SEEDS). ALL ROWS USE SAC + BC BLEND, VARYING ONLY THE TASK REWARD USED BY THE SAC COMPONENT.

Reward	Goal	Safe
Sparse only	85.3 %	98.0 %
+ Distance	95.7 %	98.1 %
+ Alignment	42.7 %	96.9 %
+ Both	72.0 %	97.7 %

is removed (SAFE-GIL: 47.3 $\rightarrow$ 36.6, CBF-RL: 36 $\rightarrow$ 32, Curriculum: 13 $\rightarrow$ 5, Reward Shaping: 43 $\rightarrow$ 13.5) or loses safety along with it (Curriculum drops to 21% safe; Reward Shaping drops to 42.7%). The headline number is the sensor-only gap between BLEND and the next-best method (SAFE-GIL): 59.1 percentage points on goal rate at a 15.8-percentage-point safety advantage. This indicates that the action-level filter labels do more than supervise safety: they encode a representation of the safe goal-directed action that survives the loss of obstacle geometry, whereas the supervision signals used by the other learning baselines do not. BLEND also outperforms PROJ on goal rate (95.7 vs. 90.3) with comparable safety, supporting the hypothesis that the smooth α -weighted label is easier for a stochastic actor to fit than the bang-bang projection.

C. Reward-Shaping Ablation

To isolate the contribution of the underlying RL signal from the contribution of the BC term, we ablate the reward function used by the SAC portion of our method. The ablation tests are run in the more challenging sensor-only environment. The result (Table III, Fig. 3) is informative: the simplest sparse reward, consisting of a one-shot goal bonus and a one-shot collision penalty, already attains 85.3% goal at 98.0% safe. Adding a per-step distance term raises goal rate to 95.7% at 98.1% safe, the strongest variant. Alignment shaping alone, which rewards the agent for facing the goal, collapses goal rate to 42.7%; combining distance and alignment partially recovers to 72.0% but does not match distance alone. Distance shaping combined with the sparse rewards is the optimal reward combination for this environment, while alignment is actively harmful. Across all four reward variants the safe rate stays within roughly 97–98%, indicating that the BC

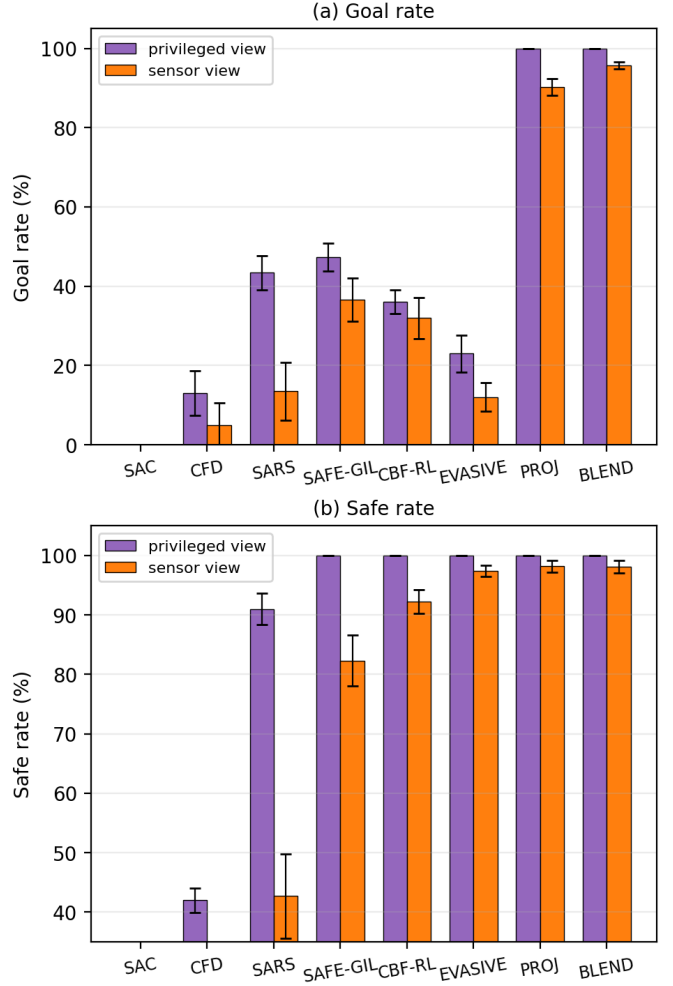


Fig. 2. Privileged-view versus sensor-only goal and safe rates per learned method on hard random. Error bars are ± 1 std across three seeds. Both SAC + BC variants retain near-privileged performance with only sensor observations, while the other learning baselines lose 10–30 percentage points on goal rate when geometry is removed.

component teaches safety reliably regardless of the RL signal; only the goal-reaching component depends on the reward.

D. Scaling with the Number of Obstacles

Training our two methods, PROJ and BLEND, at $n_{\text{obs}} \in \{3, 6, 10\}$ (Fig. 4) reveals two trends. First, safe rate is essentially insensitive to obstacle density across both methods, staying near 98 % throughout. Second, goal-reaching performance depends only mildly on density: at the sparsest setting both methods reach the goal on every scene, while at the densest setting both retain most of their goal rate. Despite navigation becoming significantly more challenging as the number of obstacles grows, the goal rate stays largely consistent, indicating that filter distillation supports the learning process effectively across difficulty levels.

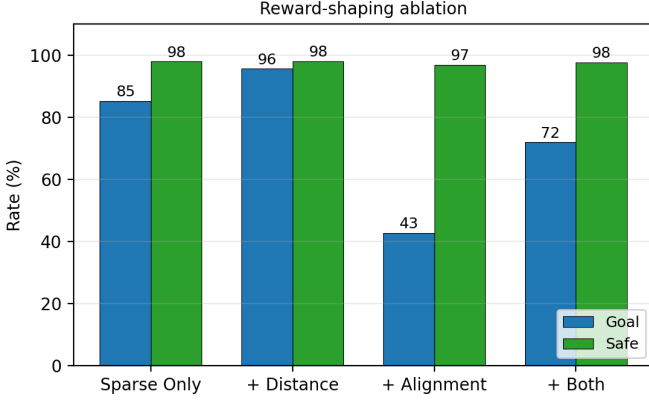


Fig. 3. Reward-shaping ablation on hard random scenarios. Sparse goal/collision terms alone already attain 85.3% goal; adding a per-step distance term yields the strongest variant at 95.7% goal; alignment shaping alone collapses goal rate to 42.7%; combining distance and alignment partially recovers to 72.0% but does not match distance alone.

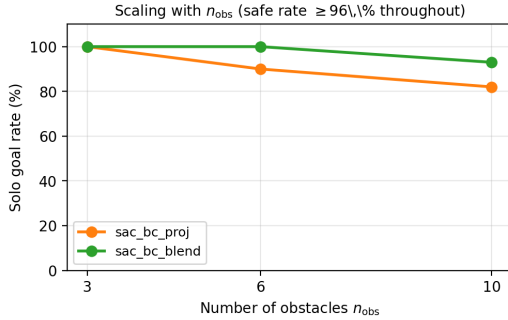


Fig. 4. Solo goal rate versus n_{obs} for the two SAC + BC variants under the best reward of §V-C. Safe rates remain high throughout. As obstacle density increases, the BC variants retain much of their goal rate with only minor degradation.

E. Sensor Robustness

Because the student observes raycast distances, a natural concern is whether our results depend on a noise-free sensor. We re-evaluate the two SAC + BC variants under Gaussian range noise $\sigma \in \{0.05, 0.10\}$ m on every ray. Both methods degrade gracefully (Tab. IV): BLEND loses about 6 pp goal rate at $\sigma=0.05$ and 12 pp at $\sigma=0.10$, while PROJ loses about 5 pp and 9 pp respectively. Safe rate degrades more visibly than goal rate, dropping from $\sim 98\%$ to $\sim 90\%$ at the larger noise level. Importantly, BLEND retains its lead over PROJ at every noise setting (e.g. 83.7 vs. 81.4 goal and 91.4 vs. 89.2 safe at $\sigma=0.10$), and even the most-degraded number (83.7% goal at $\sigma=0.10$) sits well above the noise-free goal rate of every sensor-only learning baseline in Table II. The headline numbers are therefore robust to onboard-sensor noise of the magnitude that a real lidar would produce.

F. Goal-Aware versus Goal-Blind Safety

Filter distillation is more than a mechanism for teaching a policy to be safe: it teaches the policy to be safe *and* efficient at reaching the goal. Figure 5 illustrates this on a hand-crafted

TABLE IV
SENSOR ROBUSTNESS ON HARD RANDOM UNDER GAUSSIAN RANGE NOISE (5–10 CM STD), THREE SEEDS AND 100 EPISODES PER CONDITION.

Method	noise=0	$\sigma=0.05$	$\sigma=0.10$
PROJ	90.3 / 98.2	85.5 / 91.3	81.4 / 89.2
BLEND	95.7 / 98.1	89.6 / 95.2	83.7 / 91.4

L-wall scenario. We compare a rollout from our method against a representative rollout from SAFE-GIL, a commonly used method for learning safe policies. Both SAFE-GIL and BLEND stay safe ($V_t \geq 0$) and both reach the goal, so a safety-only comparison would judge them equivalent. The trajectories, however, differ qualitatively. SAFE-GIL, whose BC supervision comes from the projected goal-seeker, behaves as a goal-blind avoider in this scenario: it routes upward into the apparently safer upper region and weaves through a long, convoluted detour before reaching the goal in 108 steps. BLEND routes downward through the bottom opening (the smooth goal-aware route the α -blend label would have produced near the boundary) and reaches the goal in 62 steps, a 43% reduction. The blend label therefore injects exactly the structural primitive that the bang-bang projection-style label cannot communicate: “remain goal-directed wherever $V > 0$, and deflect by an amount proportional to how far below the boundary the state would otherwise lie.”

G. Discussion

We see three reasons for the efficacy of our method.

(1) *The label tells the policy what to do, not just whether it is safe.* The blend label gives a specific action at every state: head toward the goal when safely inside the safe set, deflect a little near the boundary, and use the full evasive action when deep in the unsafe set. A scalar penalty cannot convey this; it only tells the agent that something went wrong, not what to do instead. Safety-Augmented Reward Shaping illustrates the limitation directly: it uses the same filter signal as our method, but only as a penalty when $V(s) \leq 0$, and reaches just 43% goal in the privileged setting and 13.5% in the sensor-only setting, compared to 100% and 95.7% for the goal-aware blend.

(2) *The blend label is smoother than what other action-level methods provide.* The other methods that do supervise actions all use a discontinuous signal of some kind. The projection label, SAFE-GIL’s expert, and CBF-RL’s regulariser all behave the same way near the boundary: they let the policy be goal-directed up to $V = 0$ and then snap to a safety override the moment the constraint is crossed. This produces a step in the supervision signal at the boundary, which a stochastic SAC actor has to fit on either side of a discontinuity. The blend label avoids the step entirely: α rises smoothly with $-V$, so the supervision transitions gradually from goal-seeking to evasive action as the state moves through the boundary. Avoiding this discontinuous step appears to help the policy learn more effectively, and to place more emphasis on goal-seeking than other methods while still maintaining safety. This

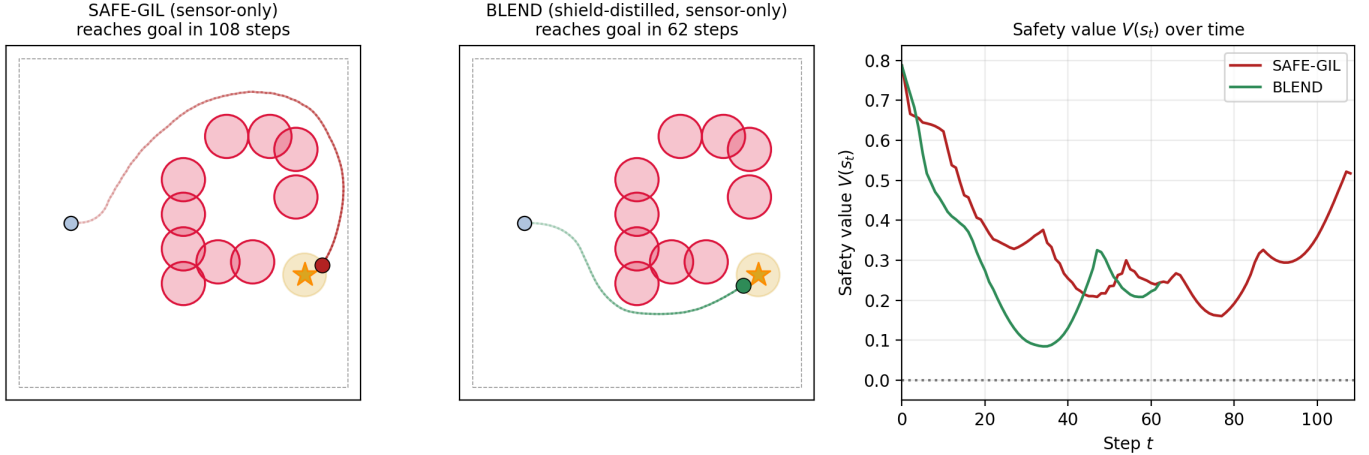


Fig. 5. Goal-aware safety from the blended distillation label on a qualitative hand-crafted L-wall scenario, sensor-only deployment, no runtime shield. The goal lies down and to the right; the L-wall blocks the direct path; the only narrow gap to the goal is below the wall. **Left:** SAFE-GIL routes upward into the open upper region and follows a long, convoluted detour, reaching the goal in 108 steps. **Middle:** BLEND (same architecture, trained on the α -blend of filter and goal-seeker actions) routes downward through the bottom opening and reaches the goal in 62 steps, a 43% reduction at matched safety. **Right:** per-step $V(s_t)$ for the two rollouts; both remain safe. The blend label adds goal direction on top of safety.

shows up in the sensor-only numbers, where BLEND reaches 95.7% goal compared to 90.3% for PROJ, 36.6% for SAFE-GIL, and 32.0% for CBF-RL, all at comparable safe rates.

(3) *Applying the BC loss only where it matters.* We turn the BC loss on only at safety-critical states. This keeps the supervision focused where the filter has something useful to say and silent everywhere else, so the SAC objective is free to discover detours without being dragged toward the BC label at every step. Applying BC at every state, as curriculum distillation does, drowns out the goal signal quickly: the Curriculum Filter Distillation baseline, which uses this approach, reaches only 13% goal at 42% safe in the privileged setting. Additionally, SAFE-GIL’s adversarial state-finding leads to over-application of filter corrections, even at states where they are not essential for learning, which limits the method’s goal performance.

VI. CONCLUSION

A safety filter is more useful as a teacher than as a runtime override. In many embodied-AI tasks, safety is part of task success; the filter contains precisely the structural information that a learned policy requires in order to be both safe and high-performing; and distilling that information into the policy raises the policy’s task return. With privileged observations, the distilled Dubins navigation policy attains 100% goal and 100% safe rates, exceeding the next-best learning baseline by 53 percentage points on goal rate. With sensor-only observations, it attains 95.7% goal and 98.1% safe, exceeding the next-best baseline by 59 percentage points on goal rate while matching or exceeding its safe rate. On a hand-crafted L-wall scenario where SAFE-GIL and the distilled student both reach the goal safely, the distilled student does so in 43% fewer steps, illustrating that the goal-aware α -blend label injects a structural primitive that scalar reward and projection-style labels cannot communicate. The distilled policy can be deployed with or without the filter; the contribution we emphasise is the

policy that respects safety as part of optimising the task, rather than the elimination of the runtime shield.

A. Future Work

Additional environments. All tests are conducted in the Dubins car environment under two observation types. Additional environments with different safety constraints would test the generality of our method.

Additional baselines. Rollout filtering [18] could not be evaluated within the scope of this work. A matched-compute head-to-head against curriculum filter distillation [1], the closest cousin of our method, would isolate the contribution of selective gating from the blended label.

Alternative blending and scaling. The linear ramp $\alpha = \text{clip}(-V/c, 0, 1)$ is a natural first choice; CBF-style blending that enforces $\dot{V} + \kappa V \geq 0$, or learned weightings on richer features, are open alternatives. Many other methods of blending could be tested to evaluate which method yields optimal performance.

Closed-form filter only. We have only studied an analytic teacher. Whether distillation is equally effective with a learned filter that has approximation error (latent CBF, DeepReach [8], or world-model-based safety) is an open question.

B. Contributions of Each Team Member

This is an individual project conducted solely by Nathan Ludlow, including all design, implementation, training, evaluation, figure generation, and writing.

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